

10 Applications of Theta Functions

The problem of the representation of an integer n as the sum of a given number k of integral squares is one of the most celebrated in the theory of numbers. Its history may be traced back to Diophantus, but begins effectively with Girard's (or Fermat's) theorem that a prime $4m + 1$ is the sum of two squares. Almost every arithmetician of note since Fermat has contributed to the solution of the problem, and it has its puzzles for us still.

G. H. Hardy, 1940

This chapter is devoted to a closer look at the theory of theta functions and some of its applications to combinatorics and number theory.

The theta function is given by the series

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z},$$

which converges for all $z \in \mathbb{C}$, and τ in the upper half-plane.

A remarkable feature of the theta function is its dual nature. When viewed as a function of z , we see it in the arena of elliptic functions, since Θ is periodic with period 1 and “quasi-period” τ . When considered as a function of τ , Θ reveals its modular nature and close connection with the partition function and the problem of representation of integers as sums of squares.

The two main tools allowing us to exploit these links are the triple-product for Θ and its transformation law. Once we have proved these theorems, we give a brief introduction to the connection with partitions, and then pass to proofs of the celebrated theorems about representation of integers as sums of two or four squares.

1 Product formula for the Jacobi theta function

In its most elaborate form, Jacobi's **theta function** is defined for $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ by

$$(1) \quad \Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}.$$

Two significant special cases (or variants) are $\theta(\tau)$ and $\vartheta(t)$, which are defined by

$$\begin{aligned} \theta(\tau) &= \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau}, \quad \tau \in \mathbb{H}, \\ \vartheta(t) &= \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}, \quad t > 0. \end{aligned}$$

In fact, the relation between these various functions is given by $\theta(\tau) = \Theta(0|\tau)$ and $\vartheta(t) = \theta(it)$, with of course, $t > 0$.

We have already encountered these functions several times. For example, in the study of the heat diffusion equation for the circle, in Chapter 4 of Book I, we found that the heat kernel was given by

$$H_t(x) = \sum_{n=-\infty}^{\infty} e^{-4\pi^2 n^2 t} e^{2\pi i n x},$$

and therefore $H_t(x) = \Theta(x|4\pi i t)$.

Another instance was the occurrence of ϑ in the study of the zeta function. In fact, we proved in Chapter 6 that the functional equation of ϑ implied that of ζ , which then led to the analytic continuation of the zeta function.

We begin our closer look at Θ as a function of z , with τ fixed, by recording its basic structural properties, which to a large extent characterize it.

Proposition 1.1 *The function Θ satisfies the following properties:*

- (i) Θ is entire in $z \in \mathbb{C}$ and holomorphic in $\tau \in \mathbb{H}$.
- (ii) $\Theta(z+1|\tau) = \Theta(z|\tau)$.
- (iii) $\Theta(z+\tau|\tau) = \Theta(z|\tau)e^{-\pi i \tau}e^{-2\pi i z}$.
- (iv) $\Theta(z|\tau) = 0$ whenever $z = 1/2 + \tau/2 + n + m\tau$ and $n, m \in \mathbb{Z}$.